

Rethinking Image Aesthetics Assessment: Models, Datasets and Benchmarks

IJCAI 2022 Summary

James Hsu - August 1th 2025

Motivation

- Standard IAA models ignore differences across image **themes**, learning aesthetic scores directly from pixel patterns
- Images of different themes (e.g. portrait, landscape, food) have **distinct evaluation criteria**
- This causes **attention dispersion** and lower model interpretability and accuracy

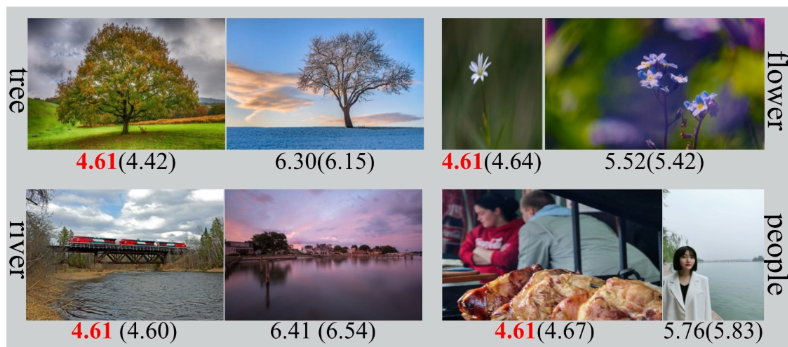


Figure 1: Examples of images and annotations in the proposed TAD66K dataset with the ground truth (and predicted) scores shown below each image. Even with the same scores, aesthetics of different themes correspond to different criteria and cannot be compared directly.

Key Contributions

- **TAD66K Dataset:** 66,000 images across 47 themes, each image annotated by 1,200+ users under theme-specific criteria
- **TANet Architecture:** Theme branch and RGB-distribution-aware attention module (RGBNet) to adaptively learn per-theme aesthetics
- **Benchmark:** Evaluation on AVA, FLICKR-AES, and TAD66K; TANet outperforms 17 prior IAA methods on all datasets

TAD66K: Theme and Aesthetics Dataset

- Contains **66K images**, manually curated across **47 popular themes** (e.g., art, nature, food)
- Each image receives at least **1,200 ratings** using **theme-specific scoring criteria**, enabling fine-grained aesthetic benchmarking

TANet Architecture Overview

- **Theme Branch (TUNet):** Extracts theme embedding and adaptive weights for guiding aesthetic evaluation
- **RGBNet Attention:** Color-distribution aware attention mechanism designed to be lightweight and theme-aligned
- **Feature Fusion:** Theme embedding modulates attention maps, focusing aesthetic prediction on relevant regions
- Achieves high accuracy with **better efficiency vs. standard self-attention**

TANet Architecture Overview

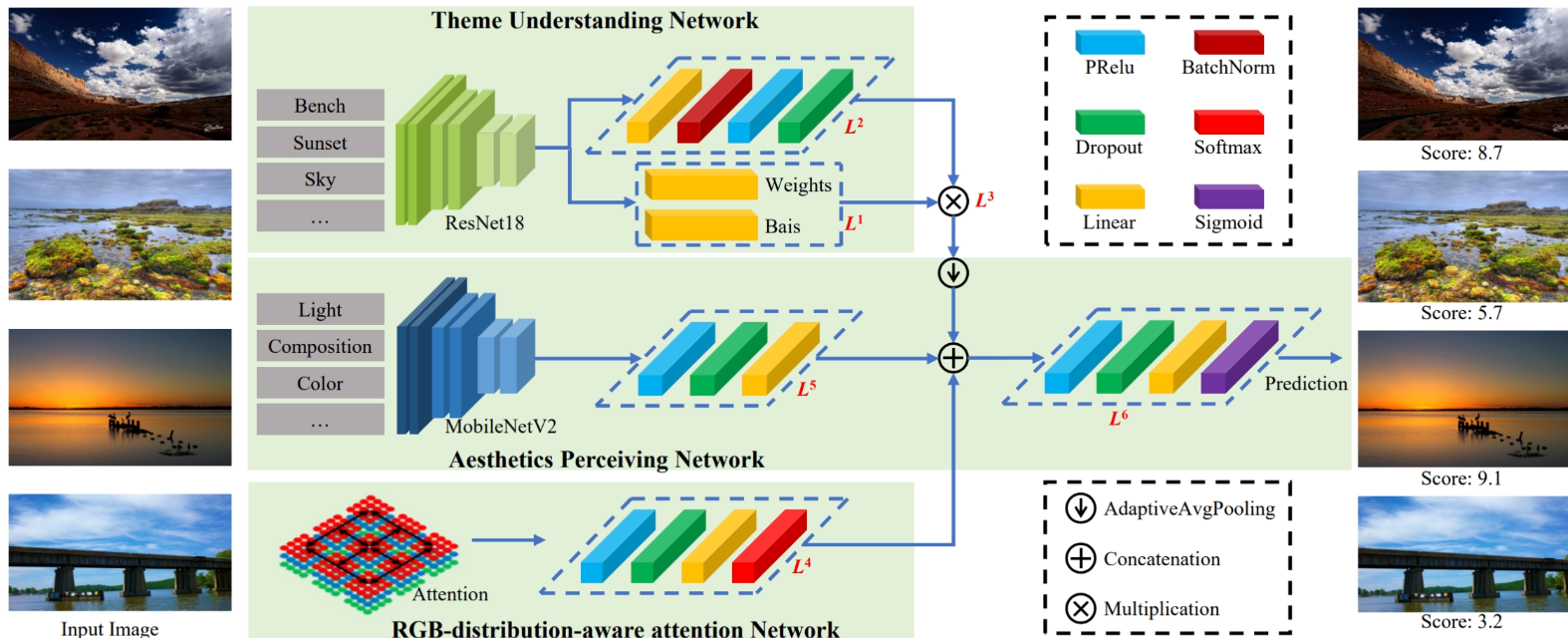


Figure 5: Overall architecture of the proposed TANet model. TANet consists of three components: TUNet is intended to efficiently learn theme information, RGBNet obtains the color similarity and distribution relationship, and APNet is introduced to fuse multiscale features to predict the aesthetic score. See section 4 for further details.

Loss Formula

$$\mathcal{L}_{\text{TANet}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

Where:

- $\hat{y}_i$ = predicted aesthetic score (continuous)
- y_i = ground-truth MOS from TAD66K
- **N** = number of training samples

Benchmark Results

- Performance on AVA, FLICKR-AES, and TAD66K compared across models
- TANet (IJCAI 2022) achieves on AVA and TAD66K respectively:
 - **SRCC $\approx$ 0.758/0.513**
 - **LCC $\approx$ 0.765/0.531**
- Outperforms prior models (e.g., MUSIQ, HLA-GCN) and establishes a new state-of-the-art baseline for IAA in 2022

Summary Table

Component	Description
Dataset (TAD66K)	66K images, 47 themes, >1,200 ratings/image
Theme Branch (TUNet)	Extracts theme embedding and adaptive perception weights
Attention Module (RGBNet)	Color-distribution-aware attention tailored to theme
Loss Training	Regression + theme classification supervision
Benchmark Metrics	SRCC $\approx$ 0.758/0.513, LCC $\approx$ 0.765/0.531 on AVA/TAD66K; beats 17 baselines

Thank you! 🙏